SMART RIDER ASSISTANCE



Table of Contents



Our Team

Background

Problem Statement

Objective

Working

USP

Innovation

Advantages

TEAM



Project Lead: Razzaq Ahamed Shajahan



Team Member: Ashwin S M

BACKGROUND

India has over **4.6 lakh*** road accidents annually, with a major share involving **two-wheelers**



NEARLY 45% of total road deaths INVOLVE **Two-wheeler users**



- Major **falate accidents** are
  - **Rider distraction** (mobile usage)
  - **One-hand** / no-hand riding
  - **Loss** of vehicle control



MAIN CAUSES OF TWO-WHEELER ACCIDENTS:

Mobile distraction
Helmets, airbags

One-hand / no-hand riding
No real-time behavioral monitoring

CURRENT SAFETY SYSTEMS

Focus on post-accident **protection**
Helmets, airbags

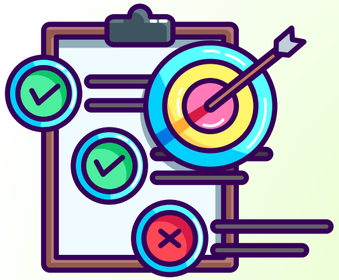
No real-time **behavioral monitoring**

PROBLEM STATEMENT



- Unsafe riding behaviors such as one-hand riding, no-hand riding, and mobile usage lead to loss of control and increased accident risk, while existing systems fail to detect and intervene in real-time.

OBJECTIVES



- To improve safety in two-wheeler riding conditions
- To identify unsafe riding behaviors in real-time
- To enhance rider awareness during critical situations
- To reduce accidents caused by human error and loss of control
- To ensure timely response to unsafe riding conditions

Smart Rider Assistance System (SRA) – Working

Real-time detection of rider's hand position on handlebars

1. **Grip Sensors**
   Continuously monitor hand position



2. **Hand Status Detected**
   Both hands?
   → Normal

3. **One/No Hand Detected**
   Unsafe condition identified



4. **Short Delay (2–3 sec)**
   Confirms unsafe condition





Multi-Stage Alerts

1. **Visual Alert**
   Warning on instrument cluster

2. **Audio Alert**
   Buzzer / beeping sound

3. **Haptic Feedback**
   Handlebar vibration (tactile alert)

Rider Response

Hands Back → System Resets

Normal Riding Continues

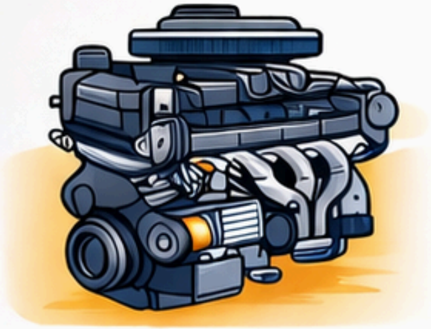
Key Highlights

- Real-time Grip Detection
- Detects Unsafe Condition
- Smart Delay Prevents False Alerts
- Multi-Stage Alerts for Awareness
- Auto Reset on Rider Response

SMART RIDER ASSISTANCE SYSTEM
SPEED REDUCTION WORKING

Automatic speed reduction process to maintain rider safety and stability

IC ENGINE VEHICLES



1 FUEL INJECTION CONTROL

Limits the fuel supply, thus reduces engine power.

2 GEAR-BASED SPEED LIMIT

Operates in lowest speed of current gear.

Vehicle operates in lower speed of current gear.

SPECIAL CASE: IDLING CONDITION

Sensor inactive in idle (800–1200 RPM)

→ Allows hands-off in traffic.



ELECTRIC VEHICLES



1 TORQUE REDUCTION CONTROL

Limits current supply to the motor, reducing torque smoothly.

EVs do not have manual gears. Speed is controlled directly through torque management.



SMOOTH & CONTROLLED SPEED REDUCTION

SMOOTH & CONTROLLED SPEED REDUCTION | **MAINTAINS RIDER STABILITY** | **PREVENTS ABRUPT BRAKING** | **AUTO RESET BRAKING**

USP

Real-time rider behavior monitoring
Intelligent safety beyond vehicle-based systems

Multi-stage alert system
Visual + audio + haptic alerts

Adaptive control
Hand-based safety intervention

Works for ICE & EV
Adaptive control

Adaptive control
Works for ICE & EV

Innovation



Grip-based detection
Safety monitoring interactions with handlebar



Context-aware logic
Adaptive control for hand position



Gear & torque control
Gear-based (ICE) + torque-based (EV)



Idle RPM intelligence
Inactive in idle traffic to avoid false alerts



Progressive safety system
Alert → Warning → Control

Adaptive control

Works for ICE & EV

Context-aware logic
Gear & torque control

Gear & torque control

SMART RIDER ASSISTANCE SYSTEM
ADVANTAGES

KEY ADVANTAGES

- **Provides early warning** before unsafe riding becomes critical
- **Prevents wheel locking** and sudden engine braking
- **Maintains vehicle stability** and balance during riding
- **Ensures smooth and controlled deceleration,** avoiding jerks
- **Encourages proper riding posture** (both hands on handlebar)



ADDITIONAL MECHANICAL ADVANTAGES

- **Reduces uneven load** on handlebar components
- **Minimizes** handlebar **misalignment** over time
- **Reduces wear** in steering and front fork components
- **Improves tyre life** by maintaining stable control
- **Enhances overall vehicle durability** and reduces **maintenance**


THANK YOU
RIDE HARD! RIDE SMART! RIDE SAFE!
RIDE
SAFE
LIVE